POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2014
Programme: BE | Full Marks: 100
Course: Microprocessor | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Describe the Von Neumann's architecture of a computer system. 5
   b) Differentiate between a microprocessor and microcontroller. 5
   c) Explain the addressing modes of Intel 8085. 5
2. a) Starting from the address $DOOO_h$ to $DOO9_h$ there is an array of ten data. Write a programme using 8085 instructions to find out how many times the data 08h repeats in the given array. Store the result at address $C000_H$. 7
   b) Write an assembly language program for Intel 8085 to compute 8

$$\sum_{i=1}^{n} \frac{x_i}{2}$$

Where X_i are three numbers stored at memory locations A001H, A002H and A003H. Store the result at memory location starting at D000H.

3. a) What is the significance of Timing Diagram? Draw the labeled timing diagram for the instruction OUT 80H. 8
   b) Draw the block diagram of Intel 8255 PPI, and explain the functions of the sub blocks. 7
4. a) Explain how two 2048*8 RAM are interfaced with 8085 microprocessor and also find the address range for each RAM chips. 7
   b) Write an 8086 ALP for MASM to find a square of a given number. 8
5. a) Draw the functional block diagram of 8086 microprocessor and compare the functions of its two basic units. 7
   b) What is Interrupt Vector Table (IVT)? Draw the IVT for 8086 microprocessor and explain different types of 8086 interrupts with respect to interrupt vector table. 8
6. a) Write an 8086 ALP for MASM to display the string "POKHARA UNIVERSITY" without using 09h on screen. Explain all steps and assume necessary data. 7
   b) Two computers are to be connected with each other via modems. Using RS-232 standards, explain the connections? Also, show the connections for null modem. 8
7. Write short notes on: **(Any two)** 2×
   a) Features of Intel 80386
   b) Synchronous & asynchronous data transmission
   c) Macros.

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